



Music

AMAZON MUSIC CLUSTERING

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▶ **Objective:** Cluster songs based on audio and metadata features.

▶ **Techniques:** Unsupervised learning using clustering algorithms.

▶ **Tools :** Python, Pandas, Scikit-learn, Matplotlib.



DATASET & PREPROCESSING

- Loaded music dataset with numerical and categorical features.
- Handled missing values and normalized data for consistency.
- Encoded categorical variables such as genre or artist info.
- Scaled features for better clustering performance.





CLUSTERING METHODOLOGY

- Applied K-Means algorithm to group songs into clusters.
- Determined optimal number of clusters using Elbow Method.
- Evaluated cluster quality using inertia and silhouette score.
- Visualized clusters to interpret song groupings.



RESULTS & CONCLUSION



Successfully grouped songs with similar characteristics.



Visualizations showed clear separation among clusters.



Insights can help in music recommendation or playlist generation.



Future work: Experiment with DBSCAN or PCA for better insights.





THANK YOU